pmsimstats Codebase Overview

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The `pmsimstats` R package implements Monte Carlo simulation-based power analysis for aggregated N-of-1 clinical trial designs, comparing four trial designs on their ability to detect a biomarker-treatment interaction. Based on Hendrickson et al. (2020), *Frontiers in Digital Health* 2:13.

1 Core R Package Files (R/)

1.1 generateData.R

The data-generating process (DGP). Two exported functions:

- **`generateData()`** – Draws simulated participant data from a multivariate normal distribution. Accepts a pre-built covariance matrix (`cached_sigma`) or builds one via `buildSigma()`. Uses

pre-computed Cholesky factor for efficient MVN draws (`L %*% Z + mu`). Computes outcome columns via vectorized `data.table` operations.

Under Architecture A (`dgp_architecture = 'mean_moderation'`), the BM-BR correlation block is omitted from Σ and the biomarker-treatment interaction is applied as an additive post-draw step: $\text{BR}_{it} \mathrel{+}= c_{bm} \cdot B_i^* \cdot \mathbf{1}[\text{on drug}] \cdot \sigma_{\text{BR}}$, where B_i^* is the standardized biomarker.

- **`buildSigma()`** – Constructs the 26x26 covariance matrix for a single trial path. Implements:
  - AR(1) within-factor autocorrelation: $\rho^{|t_i - t_j|}$
  - Cross-factor correlations (same-time and different-time with AR(1) decay)
  - Biomarker-BR correlation with exponential decay during off-drug periods: $c.bm \cdot \exp(-\lambda_{cor} \cdot tsd)$
  - Auto-derives $\lambda_{cor} = \ln(2)/\text{carryover_t1half}$
  - Positive definiteness check with verbose warning
  - Pre-computes Cholesky factor for caching
- **`validateParameterGrid()`** – Tests all parameter combinations for PD before simulation. Reports failures upfront.

1.2 generateSimulatedResults.R

The simulation loop. Single exported function:

- **`generateSimulatedResults()`** – Orchestrates the full parameter sweep:
  1. PD pre-validation of all parameter combinations
  2. Sigma matrix caching (builds all unique matrices once)
  3. Parameter grid expansion across designs, response params, baseline params, and model params
  4. Per-parameter-set worker function (`run_one_paramset`) that generates data, fits LME, applies censoring, and collects results
  5. Parallel execution via `furrr::future_map()` with closure-based function passing to workers
  6. Progressive save support for long runs

  Key parameters: `lambda_cor` (correlation decay, default NA = auto), `n_cores` (parallelism, default auto-detect).

1.3 lme_analysis.R

The analysis model. Single exported function:

- **`lme_analysis()`** – Fits a linear mixed-effects model to simulated or actual trial data:
  - Uses `nlme::lme` with `corCAR1(form = ~t|ptID)` for continuous-time AR(1) residual correlation

  - Dynamic formula selection: `Sx ~ bm + t + Dbc + bm:Dbc` when within-subject drug variation exists; `Sx ~ bm + t + bm:t` for open-label
  - `Dbc` is a continuous drug indicator (1 when on drug, exponential decay when off drug)
  - Graceful handling of rank-deficient models (returns NA)
  - Fallback to `lme` without `corCAR1` if optimization fails
  - Extracts interaction coefficient from `nlme` tTable

1.4 carryover_analysis.R

Extended analysis functions for actual trial data:

- **`characterize_carryover()`** – Sweeps candidate carryover half-lives (default 0.1 to 8 weeks), fits `nlme::lme` at each, compares AIC/BIC to identify the best-fitting half-life. Returns drug effect, interaction, AR(1) phi at each half-life.
- **`analyze_trial_extended()`** – Extracts both the main drug effect (`Dbc`) and biomarker interaction (`bm:Dbc`) with 95% CIs. Computes variance components, predicted drug effect as a function of biomarker value, and clinical decision threshold.
- **`print_carryover_summary()`** / **`print_trial_summary()`** – Clinical reporting summaries.

1.5 buildtrialdesign.R

Trial design specification:

- **`buildtrialdesign()`** – Constructs trial paths from intuitive inputs (timepoints, expectancies, on-drug vectors). Computes derived timing variables: `tod` (time on drug), `tsd` (time since discontinuation), `tpb` (time in positive belief).

1.6 utilities.R

Helper functions:

- **`modgompertz()`** – Modified Gompertz function for response trajectories. Passes through origin, asymptotes at `maxr`.
- **`cumulative()`** – Converts interval durations to cumulative time.
- **`reknitsimresults()`** – Recombines progressive save chunks.

1.7 censordata.R

Dropout simulation:

- **`censordata()`** – Two-component dropout: time-dependent (flat rate) and outcome-dependent (biased by symptom change). Implements last-observation-carried-forward censoring.

1.8 plottingfunctions.R

Visualization:

- **`PlotModelingResults()`** – Power heatmaps across 4-dimensional parameter grids with `ggplot2` `geom_tile` and `facet_grid`.
- **`plotfactortrajectories()`** – Factor trajectory plots by trial path.

2 Analysis Scripts

2.1 Figure 4 (`analysis/figure4/`)

Script	Purpose
`01-generate-power-sweep.R`	Figure 4: current code, sequential
`02-generate-power-sweep-original.R`	Figure 4: original commit code
`03-generate-power-by-commit.R`	Figure 4: parameterized by commit
`04-generate-power-head.R`	Figure 4: current code, parallel, c.bm={0,0.25,0.45}
`05-pd-diagnostics.R`	PD failure rate instrumentation
`06-plot-power-heatmaps.R`	Heatmap plotting with provenance

2.2 Figure 5 (`analysis/figure5/`)

Script	Purpose
`01-generate-parameter-sensitivity.R`	Figure 5: revised code
`02-generate-parameter-sensitivity-original.R`	Figure 5: original code

2.3 Extracted Code by Commit

Code from each historical commit extracted for comparison:

```
analysis/scripts/commit_42ac030/   # Initial (publication)
analysis/scripts/commit_8609f12/   # Ron Thomas edits
analysis/scripts/commit_f6ee86d/   # Autocorrelation typo fix
analysis/scripts/commit_f70f86d/   # Carryover added
analysis/scripts/commit_325314f/   # Modified Db (broken)
analysis/scripts/original_R/      # Copy of 42ac030
```

3 White Papers (`docs/pdf/`)

4 Bundled Data (`data/`)

5 Key Parameters

5.1 Current Configuration (Revised Code)

5.2 Publication Configuration (Original Code)

6 Dependencies

Core: `data.table`, `nlme`, `lme4`, `lmerTest`, `corpcor`, `MASS`, `ggplot2`

#	Document	Content
01	Codebase Overview	Package structure, parameters, dependencies
02	DGP Architecture Comparison	Mean moderation vs. MVN correlation
03	Audit and Revision Report	Seven DGP corrections, PD resolution, deliverables
04	Revised Power Analysis	Problems, fixes, validated results
06	AR(1) Residual Correlation	lmer to nlme transition with Type I error validation
07	Positive Definiteness Failures	PD failure prevalence and causes
08	BM-BR Correlation Decay	Pharmacokinetic derivation of λ_{cor}
09	Carryover Correlation Artifact	BM-BR correlation artifact analysis
10	Carryover Analysis Assessment	Carryover characterization in actual data
13	Figure 4 Walkthrough	Pre-audit methodology reference
14	Trial Analysis Guide	Collaborator goal implementation
15	Tidyverse Alignment Plan	Alignment between orig and 2025 pipelines
16	Tidyverse Code Review	Code review of 2025 pipeline
17	Simplification Plan	Codebase simplification strategy
18	Response Parameter Sensitivity	Response parameter sensitivity before/after
19	Mean Moderation Implementation	Implementation derivation for Architecture A

File	Content
`results_core.rda`	Published Figure 4 results (initial commit, 500 reps)
`extracted_bp.rda`	Baseline parameters (BP mean/SD, CAPS mean/SD)
`extracted_rp.rda`	Response parameters (Gompertz max/disp/rate/sd)
`CTdata.rda`	Actual clinical trial data
`results_maxes.rda`	Figure 5A results (response maxima)
`results_rates.rda`	Figure 5B results (response rates)
`results_trajectories.rda`	Factor trajectory data

Performance: `furrr`, `future` (parallel processing)

Legacy (in DESCRIPTION but used minimally): `svMisc`, `tictoc`, `ggpubr`, `gridExtra`

Parameter	Value	Rationale
Autocorrelation	AR(1), rho = 0.7	PD-feasible, clinically realistic
c.bm	{0, 0.25, 0.45}	Max PD-feasible = 0.45 (OL+BDC)
		Under Architecture A, the PD constraint on c_{bm} is removed
Carryover t_half	{0, 0.5, 1.0} weeks	Clinically realistic
lambda_cor	ln(2) / t_half (auto)	Pharmacokinetic derivation
Cross-factor (same time)	c.cf1t = 0.2	Unchanged from publication
Cross-factor (diff time)	c.cfct * $\rho^{\lvert t_i - t_j \rvert}$	AR(1) decay applied
Analysis model	nlme::lme + corCAR1	Required for AR(1) Type I error control
PD corrections	0/162 (0%)	All matrices valid

Parameter	Value	Issue
Autocorrelation	CS, rho = 0.8	PD failures at c.bm > 0.25
c.bm	{0, 0.3, 0.6}	0.6 exceeds PD-feasible range
Carryover t_half	{0, 0.1, 0.2} weeks	Too short for genuine effects
BM-BR correlation	Step function	Power drop artifact
Analysis model	lmer (no corCAR1)	Correct under CS
PD corrections	40/162 (24.7%)	Silent, uncontrolled